



Computer Organization and Assembly Language

Introduction to DOSBox

Lab 1

Student Name	Muhammad Hassan Khalid
Registration #	21L-5692
Instructor	Hazoor Ahmad
Class	CS3
Section	A1, A2, D1, D2
Semester	Fall 2022

Fast School of Computing

FAST-NU, Lahore, Pakistan

Activity 1

Assembly Language Code

```
[org 0x0100]
```

```
mov ax, 25 ; a part
```

```
mov bx, 15 ; b part
```

```
mov cx, ax
```

```
mov ax, bx
```

```
mov bx, cx
```

```
mov cx, [0x270] ; c part
```

```
mov bx, 0 ; d part
```

```
mov ax, [num1+bx]
```

```
add bx, 2
```

```
mov ax, [num1+bx]
```

```
add bx, 2
```

```
mov ax, [num1+bx]
```

```
mov ax, 0x4c00
```

```
int 0x21
```

```
num1: dw 12 , 25 , 10
```

Debugging Screenshots

AX 000A	SI 0000	CS 19F5	IP 0127	Stack +0 0000	Flags 7200
BX 0004	DI 0000	DS 19F5		+2 20CD	
CX 0000	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 0 0
CMD >					
0123 8B872C01 MOV AX,[012C+BX] 0127 B8004C MOV AX,4C00 012A CD21 INT 21 012C 0C00 OR AL,00 012E 1900 SBB [BX+SI],AX 0130 0A00 OR AL,[BX+SI] 0132 0000 ADD [BX+SI],AL 0134 0000 ADD [BX+SI],AL 0136 0000 ADD [BX+SI],AL					
1 DS:0000 CD 20 FF 9F 00 EA FF FF DS:0008 AD DE 1B 05 C5 06 00 00 DS:0010 18 01 10 01 18 01 92 01 DS:0018 01 01 01 00 02 FF FF FF DS:0020 FF FF FF FF FF FF FF FF DS:0028 FF FF FF FF EB 19 E6 11 DS:0030 A2 01 14 00 18 00 F5 19 DS:0038 FF FF FF FF 00 00 00 00 DS:0040 05 00 00 00 00 00 00 00 DS:0048 00 00 00 00 00 00 00 00					
2 DS:0000 CD 20 FF 9F 00 EA FF FF AD DE 1B 05 C5 06 00 00 DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 E6 11 DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00 DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00					
1 Step 2 ProcStep 3 Retrieve 4 Help ON 5 BRK Menu 6 7 up 8 dn 9 le 10 ri					

Activity 2

Assembly Language Code

A PART

```
[org 0x0100]

mov bx, n1
mov ax, 0
add ax, [bx]

mov bx, n2
add ax, [bx]

mov bx, n3
add ax, [bx]

mov bx, n4
add ax, [bx]

mov bx, n5
add ax, [bx]

mov [sum],ax

mov ax, 0x4c00
int 0x21

n1: dw 10
n2: dw 20
n3: dw 30
n4: dw 40
n5: dw 50
sum: dw 0
```

B PART

```
[org 0x0100]

mov bx, n
mov cx, 5
mov ax, 0

l1: add ax, [bx]
```

```

add bx,2
sub cx,1
jnz l1

mov [sum],ax

mov ax, 0x4c00
int 0x21

n: dw 10,20,30,40,50
sum: dw 0

```

Debugging Screenshots

PART A

AX 0096	SI 0000	CS 19F5	IP 011F	Stack +0 0000	Flags 7204
BX 012C	DI 0000	DS 19F5		+2 20CD	
CX 0030	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 0 0 1 0

CMD >				1 0 1 2 3 4 5 6 7															
011C A32E01 MOV [012E],AX				DS:0000 CD 20 FF 9F 00 EA F0 FE															
011F BB004C MOV AX,4C00				DS:0008 AD DE 1B 05 C5 06 00 00															
0122 CD21 INT 21				DS:0010 18 01 10 01 18 01 92 01															
0124 0A00 OR AL,[BX+SI]				DS:0018 01 01 01 00 02 FF FF FF															
0126 1400 ADC AL,00				DS:0020 FF FF FF FF FF FF FF FF															
0128 1E PUSH DS				DS:0028 FF FF FF FF EB 19 C0 11															
0129 0028 ADD [BX+SI],CH				DS:0030 A2 01 14 00 18 00 F5 19															
012B 0032 ADD [BP+SI],DH				DS:0038 FF FF FF FF 00 00 00 00															
012D 00960000 ADD [0000+BP],DL				DS:0040 05 00 00 00 00 00 00 00															
				DS:0048 00 00 00 00 00 00 00 00															

2	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
DS:0000	CD	20	FF	9F	00	EA	F0	FE	AD	DE	1B	05	C5	06	00	00	= f.Ω≡■ i ..†...
DS:0010	18	01	10	01	18	01	92	01	01	01	00	02	FF	FF	FF	FF	ff.
DS:0020	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	EB	19	C0	11	δ.L.
DS:0030	A2	01	14	00	18	00	F5	19	FF	FF	FF	FF	00	00	00	00	ó.....J.
DS:0040	05	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	

1 Step	2 ProcStep	3 Retrieve	4 Help ON	5 BRK Menu	6	7 up	8 dn	9 le	10 ri
--------	------------	------------	-----------	------------	---	------	------	------	-------

AX 0096	SI 0000	CS 19F5	IP 0118	Stack +0 0000	Flags 7244
BX 0127	DI 0000	DS 19F5		+2 20CD	
CX 0000	BP 0000	ES 19F5	HS 19F5	+4 9FFF	OF DF IF SF ZF AF PF CF
DX 0000	SP FFFE	SS 19F5	FS 19F5	+6 EA00	0 0 1 0 1 0 1 0

CMD >				1	0	1	2	3	4	5	6	7
0115 A32701	MOV	[0127],AX	DS:0000	CD	20	FF	9F	00	EA	F0	FE	
0118 B8004C	MOV	AX,4C00	DS:0008	AD	DE	1B	05	C5	06	00	00	
011B CD21	INT	21	DS:0010	18	01	10	01	18	01	92	01	
011D 0A00	OR	AL,[BX+SI]	DS:0018	01	01	01	00	02	FF	FF	FF	
011F 1400	ADC	AL,00	DS:0020	FF	FF	FF	FF	FF	FF	FF	FF	
0121 1E	PUSH	DS	DS:0028	FF	FF	FF	FF	EB	19	C0	11	
0122 0028	ADD	[BX+SI],CH	DS:0030	A2	01	14	00	18	00	F5	19	
0124 0032	ADD	[BP+SI],DH	DS:0038	FF	FF	FF	FF	00	00	00	00	
0126 00960000	ADD	[0000+BP],DL	DS:0040	05	00	00	00	00	00	00	00	
			DS:0048	00	00	00	00	00	00	00	00	

2	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
DS:0000	CD	20	FF	9F	00	EA	F0	FE	AD	DE	1B	05	C5	06	00	00
DS:0010	18	01	10	01	18	01	92	01	01	01	01	00	02	FF	FF	FF
DS:0020	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	EB	19	C0	11
DS:0030	A2	01	14	00	18	00	F5	19	FF	FF	FF	FF	00	00	00	00
DS:0040	05	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

1 Step	2 ProcStep	3 Retrieve	4 Help ON	5 BRK Menu	6	7 up	8 dn	9 le	10 ri
--------	------------	------------	-----------	------------	---	------	------	------	-------

Activity 3

Assembly Language Code

```
[org 0x0100]
```

```
mov ax, 200  
mov bx, 150  
mov byte[250], 50  
mov byte[200], 25
```

```
mov bx, 200  
mov ax, [bx]
```

```
mov cx, [250]
```

```
mov ax, 0x4c00  
int 0x21
```

Debugging Screenshots

AX	0019	SI	0000	CS	19F5	IP	0119	Stack	+0	0000	Flags	7200
BX	00C8	DI	0000	DS	19F5				+2	20CD		
CX	0032	BP	0000	ES	19F5	HS	19F5		+4	9FFF	OF	DF
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6	EA00	IF	SF
											ZF	AF
											PF	CF
											0	0
											1	0
											0	0
											0	0

CMD >				1	0	1	2	3	4	5	6	7
0115	8B0EFA00	MOV	CX,[00FA]	DS:0000	CD	20	FF	9F	00	EA	FF	FF
0119	B8004C	MOV	AX,4C00	DS:0008	AD	DE	1B	05	C5	06	00	00
011C	CD21	INT	21	DS:0010	18	01	10	01	18	01	92	01
011E	0000	ADD	[BX+SI],AL	DS:0018	01	01	01	00	02	FF	FF	FF
0120	0000	ADD	[BX+SI],AL	DS:0020	FF	FF	FF	FF	FF	FF	FF	FF
0122	0000	ADD	[BX+SI],AL	DS:0028	FF	FF	FF	FF	EB	19	E6	11
0124	0000	ADD	[BX+SI],AL	DS:0030	A2	01	14	00	18	00	F5	19
0126	0000	ADD	[BX+SI],AL	DS:0038	FF	FF	FF	FF	00	00	00	00
0128	0000	ADD	[BX+SI],AL	DS:0040	05	00	00	00	00	00	00	00
				DS:0048	00	00	00	00	00	00	00	00

2	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
DS:0000	CD	20	FF	9F	00	EA	FF	FF	AD	DE	1B	05	C5	06	00	00
DS:0010	18	01	10	01	18	01	92	01	01	01	01	00	02	FF	FF	FF
DS:0020	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	FF	EB	19	E6	11
DS:0030	A2	01	14	00	18	00	F5	19	FF	FF	FF	FF	00	00	00	00
DS:0040	05	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
= f.Ω i ..†...																
.....ff.																
δ.μ.																
ó.....J.																
.....																

1	Step	2	ProcStep	3	Retrieve	4	Help ON	5	BRK Menu	6		7	up	8	dn	9	le	10	ri
---	------	---	----------	---	----------	---	---------	---	----------	---	--	---	----	---	----	---	----	----	----

Activity 4

Assembly Language Code

```
[org 0x0100]

mov si,0
mov cx,12

mov bx, array
l1: add ax, [bx+si]
add si,2
sub cx,1
jnz l1

mov [sum],ax

mov ax, 0x4c00
int 0x21

array: dw 111,999,888,888,11,99,88,88,1,9,8,8
sum : dw 0
```

Debugging Screenshots

AX 0C7E SI 0018 CS 19F5 IP 0118 Stack +0 0000 Flags 7244

BX 011D DI 0000 DS 19F5 +2 20CD

CX 0000 BP 0000 ES 19F5 HS 19F5 +4 9FFF 0F DF IF SF ZF AF PF CF

DX 0000 SP FFFE SS 19F5 FS 19F5 +6 EA00 0 0 1 0 1 0 1 0

CMD >

0115 A33501 MOV [0135],AX

0118 B8004C MOV AX,4C00

011B CD21 INT 21

011D 6F OUTSW

011E 00E7 ADD BH,AH

0120 037803 ADD DI,[BX+SI+03]

0123 7803 JS 0128

0125 0B00 OR AX,[BX+SI]

0127 6300 ARPL [BX+SI],AX

1

0 1 2 3 4 5 6 7

DS:0000 CD 20 FF 9F 00 EA F0 FE

DS:0008 AD DE 1B 05 C5 06 00 00

DS:0010 18 01 10 01 18 01 92 01

DS:0018 01 01 01 00 02 FF FF FF

DS:0020 FF FF FF FF FF FF FF FF

DS:0028 FF FF FF FF EB 19 C0 11

DS:0030 A2 01 14 00 18 00 F5 19

DS:0038 FF FF FF FF 00 00 00 00

DS:0040 05 00 00 00 00 00 00 00

DS:0048 00 00 00 00 00 00 00 00

2

0 1 2 3 4 5 6 7 8 9 A B C D E F

DS:0000 CD 20 FF 9F 00 EA F0 FE AD DE 1B 05 C5 06 00 00

DS:0010 18 01 10 01 18 01 92 01 01 01 01 00 02 FF FF FF

DS:0020 FF FF FF FF FF FF FF FF FF FF FF FF EB 19 C0 11

DS:0030 A2 01 14 00 18 00 F5 19 FF FF FF FF 00 00 00 00

DS:0040 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00

= f.Ω≡■ ¡|..†...

.....ff.

δ.L.

ó.....J.

.....

1 Step 2ProcStep 3Retrieve 4Help ON 5BRK Menu 6

7 up 8 dn 9 le 10 ri